

Referee package (DRAFT) – near-gap residual closure with explicit margins, and the self-caught $O(I^2)$ correction

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

1. Purpose and scope

This referee document concerns the NEAR-GAP residuals of B1-RH-ENUM: convention exactness with explicit margins near the Brazovskii gap, and the honest retraction of an earlier $O(I^2)$ over-claim. **Scope:** the near-gap convention and the common-mode repair. **Not claimed:** the superseded $O(I^2)$ numbers are withdrawn (recorded below).

2. The result

(i) The earlier R-U10-1 $O(I^2)$ numerical claims are SUPERSEDED by the u -series triage: the remainder is 1.65×10^{-3} (not 2.7×10^{-5}), the protection is $\times 2$ (not $\times 130$), and the identity is $M' = -J(0)/2$ (not $-J(0)$). (ii) With the corrected convention, the near-gap residual closes with explicit margins via the common-mode repair.

3. Structure of the argument

The near-gap common-mode is repaired and the convention re-derived to exactness; the corrected $M' = -J(0)/2$ identity (a factor-2 convention fix) propagates to the protection ratio. The earlier over-claim is explicitly retracted and the corrected margins recorded – a self-caught audit.

4. Numerical reproduction

The common-mode repair script reproduces the corrected remainder 1.65×10^{-3} and the $\times 2$ protection; run it from the bundle.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B1-RH-ENUM / near-gap referee package (DRAFT)
Precise statement:	R-U10-1's $O(I^2)$ claims are SUPERSEDED by useries-triage (remainder $1.65e-3$ not $2.7e-5$; protection x2 not x130; identity $M'=-J(0)/2$ not $-J(0)$); with the corrected convention the near-gap residual closes with explicit margins via the common-mode repair.
Scope:	near-gap convention exactness + common-mode repair; includes a self-caught retraction.
Evidence grade:	convention-exact closure + retraction record (EXECUTED).
Reproduction command:	python codes/vacuum/neargap_common_mode_repair.py
Falsifier:	a near-gap configuration where the corrected margins fail; or a re-derivation contradicting $M'=-J(0)/2$.
Tier before / after:	DRAFT referee package; convention closure + retraction. NOT operator-confirmed.
No-overclaim statement:	the corrected (factor-2) convention is the claim; the prior $O(I^2)$ numbers are withdrawn.
Next required action:	operator review + confirm -> PUBLISHED bundle.